

Saniya

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SUMMARY

Master's student in Computer Science at Blekinge Tekniska Högskola, specialising in Artificial Intelligence and Machine Learning, with experience in designing and evaluating computer vision pipelines using CNN, TensorFlow and OpenCV. Proficient in Python for ML development and experienced in full-stack web development, including REST API integration and frontend-backend systems using JavaScript and HTML. Comfortable working across the full project lifecycle from data collection and model development to deployment and documentation both independently and as a team. Brings a strong academic foundation in core CS, a genuine interest in applying AI and software engineering challenges and a proven ability to deliver quality work under pressure.

EDUCATION

Master of Science in Computer Science Jan 2026 - Jun 2027
Blekinge Tekniska Högskola (BTH), Karlskrona, Sweden

B.Tech in Computer Science and Engineering Graduated, 2025
Jawaharlal Nehru Technological University (JNTUH) | CGPA: 9.15/10

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML, C, R

Web & Full-Stack: HTML, JavaScript, REST APIs, CSV based backends, frontend-backend integration

AI & ML: Supervised & Unsupervised Learning, CNN, TensorFlow, OpenCV, scikit-learn

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, OOP

Databases: SQL, DBMS, Information Retrieval System

DevOps & Tools: Git, GitHub, Docker, VS Code, Android Studio, Power BI, Tableau, Cryptool, Wireshark

Other: Cyber Security, Mobile Development (Android), Technical Documentation, Cisco Packet Tracer

SOFT SKILLS

Teamwork & Collaboration | Effective Communication | Adaptability & Flexibility

Attention to Detail | Analytical Thinking & Problem Solving | Strong Work Ethic

PROJECTS

AI-Integrated Multi-Sensor Crane Intelligence System | *Python, PPO, LSTM*, PEMA Student Challenge 2025/26

- Identified fundamental bottlenecks in Ship-to-Shore crane operations and designed a rigorous nine-tier AI architecture integrating PPO-based anti-sway control, PointNet++ for pose estimation, YOLOv9+OCR for container identification and LSTM-based structural monitoring on an NVIDIA Jetson Orin Nano.
- Applied structured empirical evaluation including sensitivity analysis across operating scenarios and quantitative ROI modelling, ensuring reproducibility and robustness.
- Produced comprehensive documentation covering architecture specifications and performance projections, resulting in a projected **30-40% reduction** in crane cycle time.

Face Recognition Based Attendance System | *Python, CNN, TensorFlow, OpenCV, scikit-learn*

- Designed and built a fully automated, real-time attendance system using CNN-based facial recognition, delivering a production-ready application with **92% accuracy** under varied lighting and angle conditions.
- Integrated OpenCV for live video processing and scikit-learn for model evaluation; reduced manual attendance effort by **80%** and eliminated human error through end-to-end pipeline automation.
- Built a scalable Python backend for structured data storage and retrieval, ensuring the system was ready for classroom or institutional deployment with minimal configuration.

Musical Style Modelling via Machine Learning | *Python, Supervised Learning, Feature Extraction*

- Developed and trained ML models to classify musical styles from raw audio data; achieved **85% accuracy** across **5 style categories** using MFCCs, spectral features and supervised learning algorithms.
- Led a **4-member team** through the full project lifecycle from data collection and preprocessing to model development and final academic presentation ensuring quality at every stage.
- Evaluated model performance rigorously using precision, recall and F1 metrics; iterated on results systematically and documented findings clearly for academic submission.

Dynamic Indexing Allocation | *Python, Machine Learning, Linear Regression, SQL*

- Built an intelligent data indexing system using ML and linear regression that improved database retrieval speed by 25%, validated across 3 simulated industrial use cases including logistics and inventory.

- Designed and stress tested the dynamic allocation algorithm during a hackathon, delivering a fully working solution under strict time pressure and competitive conditions.
- Wrote modular, well commented code to ensure the solution was maintainable, extensible and easy to hand over reflecting a professional approach to software quality.

Travel Chatbot | *Python, JavaScript, HTML, CSV*

- Designed and built a full-stack tourism chatbot covering 50+ destinations, integrating a Python, CSV backend with an HTML, JavaScript frontend to deliver a smooth, accessible user experience.
- Reduced user search time by 40% through structured, intelligent conversation flows that surface relevant information quickly prioritising the end user experience throughout the development process.
- Handled the entire development cycle independently, from system design to frontend integration, demonstrating ownership and the ability to deliver a complete product alone.

Picture Quiz Application | *Java, Java Swing*

- Implemented dynamic questioning and scoring mechanisms, improving user engagement and application flow.
- Co-developed an interactive quiz application in Java Swing; improved UI response time by **30%** by refactoring rendering logic and optimising event handling within the Swing framework.
- Contributed to component design, integration testing and deployment as part of a collaborative team, maintaining high code quality and performance.

EXPERIENCE & HACKATHONS

Hackxelerate Hackathon : Microsoft

CBIT, Hyderabad, India (2023)

- Competed in a **24-hour Microsoft hackathon**; delivered a functional prototype under strict time constraints and secured a position among the Top 50 teams out of 200+ participants, demonstrating the ability to produce quality output under pressure.

Algorand Blockchain Hackathon

Delhi, India (2024)

- Participated in a national-level blockchain hackathon; implemented solutions using Algorand smart contracts and gained practical experience in decentralised application architecture.

LEADERSHIP & VOLUNTEERING

- Served as **Lead of the Registration and Tech Committee** for the **JNTUH Global Alumni Meet 2025**, managing end-to-end operations for an event attended by **600+ alumni** and coordinating **50+ volunteers**; demonstrated ownership, cross-team communication, and calm delivery under pressure.
- Served as **Major Project Lead**, guiding a 4-member team through the complete lifecycle of an ML-based research project from ideation and data collection to model development and final academic presentation (2024-2025).
- Volunteered across Hackathon organisation, Job Mela (2 years), and JNTUH Global Alumni Meet 2023, actively contributing to event coordination, participant management, and community engagement (2022-2024).

CLUBS & SOCIETIES

Coding Club | Algorand Blockchain Club | Literary Club | Photography Club

ACHIEVEMENTS

- **Merit Scholarship - 50% Tuition Fee Waiver:** Awarded a full 50% tuition fee waiver throughout the entire B.Tech program in recognition of consistently outstanding academic performance. This scholarship was merit-based and renewed each year, reflecting sustained dedication and academic discipline across all four years of study.
- **1st Place : E-Raksha Artcade Competition** (CyberPeace & NCERT, National Level): Secured first position in a nationally organised cyber awareness competition jointly conducted by CyberPeace Foundation and NCERT. The competition required participants to communicate complex cybersecurity concepts through visual design, combining technical understanding with creative communication, a skill directly relevant to client-facing work.
- **Appreciation Certificate : JNTUH Global Alumni Meet 2023** (Directorate of Alumni Affairs, JNTUH): Recognised by the university's Directorate of Alumni Affairs for outstanding contribution as a volunteer at the Global Alumni Meet 2023, where responsibilities included participant coordination, technical support, and on-ground event management.
- **Appreciation Certificate : Mega Job Fair** (JNTUH, Sewa International & K. Reddy P. Reddy Foundation): Received formal recognition for dedicated service and coordination support at a large-scale job fair organised in collaboration with Sewa International, connecting hundreds of students with employment opportunities.
- **Participation : Hackxelerate Hackathon** (Microsoft, CBIT Hyderabad): Competed in a 24-hour Microsoft-organised hackathon and delivered a functional prototype, finishing among the Top 50 out of 200+ teams.

LANGUAGES

English : Full Professional Proficiency | **Hindi** : Native | **Swedish** : Beginner level